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HIND OIL INDUSTRIES: DEMAND ANALYSIS

Abhishek Rohit, Debdatta Pal, and Pradyumna Dash wrote this case solely to provide material for class discussion. The authors do not intend to illustrate either effective or ineffective handling of a managerial situation. The authors may have disguised certain names and other identifying information to protect confidentiality.

The digital clock blinked 2:40 a.m. as Abhishek Khemka, the manager of Hind Oil Industries (HOI), tossed and turned in his bed on a rainy September night in 2015. The next day at breakfast, he approached his father, Ashok Khemka, the veteran businessman who had founded HOI, an edible mustard oil manufacturing and selling unit based in the Indian city of Asansol.[1]

Abhishek had joined HOI as an assistant in 2011 after earning a master of business administration degree with a focus on finance. After gaining an in-depth understanding of the business affairs, he took over from his father in March, 2014. Nevertheless, he continued to consult his father on every critical issue.

"At this rate we will be running losses in a few weeks' time. Our price is too low," Abhishek blurted out. "The price of mustard seeds has skyrocketed. Our price cannot absorb the high production costs any longer."[2] Ashok looked at his son thoughtfully and remarked, "This is the third time in a week that you have brought this up. We have always sold our product at a price considerably lower than that of the market. That has been our strategy from the beginning. Who will buy our product if we price it so close to that of major national brands?"

"We are priced too low," Abhishek repeated. "Our brand has earned the reputation of being a quality product for all these years and it's time we leveraged that reputation in the present situation. I disagree that an increase in price will affect our revenue adversely." Abhishek paused for a moment and then concluded, "I shall get back to you after some calculations."

Ever since Abhishek had joined the business, he had meticulously maintained monthly records of all the business transactions in a spreadsheet. He regularly used this along with the tools and techniques he had learned in college to take business decisions. However, the current dilemma was new to him, and he wondered if he could combine some of the theories of microeconomics with demand-modelling techniques to resolve the issue.

[1] Asansol was the second-largest city in West Bengal after Kolkata and was ranked the 42nd-fastest-growing city in the world by city mayors in 2010.
[2] The cost of mustard seeds accounted for around 75 per cent of the total production costs of Maa mustard oil.

HIND OIL INDUSTRIES

HOI was incorporated in July 1992 with a single oil-processing machine. It sold its product, Maa mustard oil, to local consumers in Asansol. HOI focused solely on edible mustard oil, which was extensively used as a cooking medium in eastern India. Owing to its focus on providing high-quality mustard oil, HOI established a reputation for purity with time. This allowed the company to steadily grow and expand the scale of its operations. In 2001, it became the first mill in Asansol to produce small sachets of edible oil, and this strengthened its foothold in the retail segment. Thereafter, the company operated with three processing machines and gradually established its presence in the entire Asansol market and nearby suburban areas.

Maa mustard oil was primarily sold through retail outlets to household consumers. Bulk consumers, including restaurants, food service companies, and caterers, either contacted the company directly for discounted deals or negotiated through brokers. Abhishek explained:

> Around 60 per cent of our sales are through retail stores. Households do not switch brands frequently in this product category as the taste and signature pungent smell of mustard oil varies across brands. The same does not hold true for bulk consumers, who are always looking for discounted deals.

The sales were seasonal in nature as well. There was a significant hike in the sales in certain months owing to festivals and marriage ceremonies.[3]

EDIBLE OIL MARKET IN INDIA

The edible oil market in India was very diverse and fragmented. The consumption pattern of edible oils across the country varied depending on the oil seeds that were cultivated in each particular agro-climatic zone. The indigenous edible oil industry in India was primarily dependent on the production of soybeans, groundnuts, and rapeseeds, which accounted for 88 per cent of the total oil seeds produced. However, the cultivation of oil seeds and the manufacturing of indigenous edible oils lagged far behind the demand for edible oil. This demand gap was met by imports, which rose from 38.95 per cent of edible oil consumption in 2004–2005 to 67.33 per cent in 2014–2015. More importantly, cheap palm oil and palm oil blends accounted for about 54 per cent of the crude edible oil imported in India. India was the largest importer of palm oil and soybean oil in the world in 2015.[4]

In terms of market share, palm oil, soybean oil, and mustard oil represented 42 per cent, 17 per cent, and 13 per cent of the edible oil market in India, respectively. Imported palm oil dominated the edible oil market because of its cheap price, which was supported by a falling cost, insurance, and freight price and a reduction in import duties (see Exhibit 1). This had stiffened the competition in the price-sensitive market and had made small localized oil manufacturers unprofitable.

New varieties of oil, including cottonseed oil, sunflower oil, rice bran oil, and olive oil, had also entered the market in recent years. While the market for the edible oil industry in India was heavily fragmented, with competitors struggling to grab marginal shares, it was always exposed to threats from new entrants as

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[3] In India, Hindu marriages were held on auspicious marriage dates, which were based on the Hindu calendar, *Panchang*.
[4] Credit Analysis & Research Limited (CARE Ratings), "Outlook of Indian Edible Oil Industry," Care Ratings, accessed October 25, 2016, www.careratings.com/upload/NewsFiles/SplAnalysis/Outlook%20of%20Indian%20Edible%20Oil%20Industry.pdf.

it was underpenetrated. The level of per capita edible oil consumption in India was 14.4 kilograms (kg) per year—far lower than the global average of 24 kg per year.[5] Population expansion and a rise in per capita income were fuelling an increasing demand for edible oil, making it a lucrative industry. Consumers' rising income levels led to a gradual shift in preference toward healthier refined oils and healthier imported oils like canola, olive, and castor oils, and this affected the market shares of various edible oils.

The edible oil industry in India had two main categories: an organized branded segment, which sold packaged oil, and an unorganized segment, which sold loose oil. The unorganized segment was dominant and represented around 75 per cent of the total market.[6] However, a rise in income levels and fears of adulteration—blending of edible oils in undesirable proportions—meant that consumers were gradually shifting their preferences toward packaged branded oil.[7] This had drawn several large new companies into the organized branded market in the recent years.

The consumers of edible oil included a household retail segment—those whose monthly consumption was under 5 kg—and bulk consumers, who purchased and consumed edible oil in bulk. Consumers in the household retail segment had diverse tastes and preferences, which were influenced by geographical location and income level. Location was important, as consumers became accustomed to the taste of the edible oils that were primarily produced in their region. For example, coconut was extensively cultivated and easily available in the southern part of India, and this was responsible for the use of coconut oil as the primary edible oil in the region. Similarly, mustard seeds were cultivated in the north, and mustard oil was primarily consumed in the eastern and northern parts of the country.

However, income levels also had an important role to play. Consumers in higher income groups preferred healthier oils with a high smoke point,[8] like refined oils. Even if they consumed indigenously produced edible oils, consumers preferred premium brands. Consumers with lower income levels preferred loose edible oils or non-premium packaged brands that were manufactured locally because of the price benefits. In the last few years, easy availability of cheaper imported oils such as palm oil had hurt the market share of local non-premium packaged brands such as HOI, as lower income consumers were shifting away from non-premium packaged indigenous oils to packaged palm oils, which were quite cheap.

MUSTARD OIL MARKET IN INDIA

Mustard seeds were grown only in the *rabi* (winter) season and were harvested between February and April every year. Thus, the market received its fresh stock in those months. Mustard oil was extensively consumed in northern, eastern, and north-eastern India as an edible oil for daily consumption and making pickles, and as a skin and hair oil. Mustard oil consumption in India was growing at a rate of around 20 per cent every year.[9]
Mustard oil processing in India was mainly an unorganized business; there were around 7,000–9,000 manufacturing units, and only 20 per cent of these units were registered.[10] However, the number of

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[5] Ibid.
[6] Prasenjit Chakraborty, "Strategising Growth Through Brand Acquisition," July, 2013, accessed November 27, 2016, www.mrssindia.com/media/data/strategy-mncs-in-edible-oil-industry-july-2013.pdf.
[7] Umesh Isalkarl, "FDA Seizes Stock of Adulterated Soybean, Sunflower Oil," May 28, 2015, accessed November 27, 2016, http://timesofindia.indiatimes.com/city/pune/FDA-seizes-stock-of-adulterated-soybean-sunflower-oil/articleshow/47451097.cms.
[8] The smoke point was the temperature at which cooking oil began to break down when heated. Edible oils with higher smoke points were healthier as cooking oil.
[9] Hema Yadav, Sayed Kokab Habeeb, and Manjushree Deshpande, *The Hindu Business Line*, "3 Ways to Promote Mustard Oil," January 19, 2015, accessed November 27, 2016, www.thehindubusinessline.com/markets/commodities/3-ways-to-promote-mustard-oil/article6802273.ece.
[10] Ibid.

processing units had gone down over the years. Many small manufacturers had been pushed out of the market due to unprofitability caused by the emergence of large companies, increasing competition from cheaper imported palm oil, and changing tastes and preferences among consumers. Small mustard oil manufacturers used mechanical crushing processors to extract oil from oil seeds. This was an inefficient process that led to a loss of around 10 per cent of the edible oil during extraction. The by-products of mustard oil production were used as cattle feed. Larger companies, which spent heavily on technological innovation and modernization of their extraction process, were at a considerable advantage.

The mustard oil industry as a whole was not significantly threatened by the factors that affected other edible oil varieties. This was because of the strong affinity of consumers toward the pungency of mustard oil, which was greatly appreciated as a taste enhancer and an appetite stimulant and could not be substituted by other edible oils. Its use for multiple other purposes also made it difficult to substitute. However, the expansion of mustard oil consumption remained limited to certain areas of the country as its pungency made it unacceptable in various other areas. Mustard oil was not suitable as an export because its high content of erucic acid was found to have an adverse impact on health.[11] (The U.S. Food and Drug Administration required all mustard oil to be labelled "for external use only.") India's National Institute of Agricultural Marketing created a strategy to reduce the erucic acid content and viscosity of the oil in order to make it more acceptable to consumers around the country and worldwide. However, the mustard oil industry remained a largely unorganized business serving consumers with low income levels in rural and semi-urban areas. Mustard oil thus remained positioned as a poor person's oil.

THE MUSTARD OIL MARKET IN ASANSOL

In Asansol, the number of active mustard oil-processing mills had been on a steep decline since 2005. HOI was one of the 20–25 mustard oil mills operating in Asansol that sold packaged mustard oil and were able to survive the ever-increasing competition. The West Bengal state of India, where Asansol was located, accounted for a third of the entire ₹3 billion[12] mustard oil market in the country,[13] because mustard oil was indispensable for Bengali cuisine. Therefore, the threats from other edible oils were not too pronounced in Asansol.

However, the mustard oil industry had strong internal competition. The significant rise in per capita income in West Bengal (see Exhibit 2) was reshaping consumers' preferences; there was a shift in demand away from loose oils and toward packaged oils and away from non-premium local varieties toward premium varieties. Another important factor in this shift was the rampant adulteration that was prevalent in the loose mustard oil segment. Such a shift was less conspicuous for bulk consumers, who were lured by the cheaper price of loose mustard oil. HOI had closed its operations in the loose oil segment and had shifted completely to selling packaged Maa mustard oil since 2009. This helped the company protect its reputation in Asansol as a pure quality mustard oil brand, but it had hurt its bulk sales significantly, as loose oil was priced around 8–10 per cent lower than the packaged variety. However, household retail consumers welcomed this move, and overall sales did not drop much.

[11] Amy Zavatto, "Essential Oil? The Controversy over Mustard Oil," FoxNews, January 23, 2012, accessed November 17, 2016, www.foxnews.com/food-drink/2012/01/23/essential-oil-controversy-over-mustard-oil.html.
[12] ₹ = INR = Indian rupee; all currency amounts are in ₹ unless otherwise specified; US$1.00 = ₹65.57 on September 30, 2015.
[13] Abhishek Law and Shobha Roy, "Ruchi Soya's New Recipe to Market Mustard Oil Brand," *The Hindu Business Line,* February 4, 2013, accessed October 28, 2016, www.thehindubusinessline.com/companies/ruchi-soyas-new-recipe-to-market-mustard-oil-brand/article4378701.ece.

This sole focus on the packaged and branded oil segment proved to be a double-edged sword for HOI. Not only did it lower the demand from the bulk consumers, it also exposed HOI to intense competition from the large companies that dominated the branded and packaged segment and reaped extensive benefits from economies of scale. These companies were highly efficient in oil extraction, used advanced technologies for oil processing, and extensively promoted their brands through television and other media. They also charged a premium price for their brands (see Exhibit 3) compared to the prices of small local manufacturers like HOI. Small companies like HOI, which had limited resources, spent meagre amounts on promotion (see Exhibit 4) in the form of small printed posters, which were attached to the walls of ration shops in semi-urban and rural areas. This was an affordable option for HOI, and it seemed helpful in retaining existing consumers and penetrating deeper into newer rural markets.

Problems were significantly aggravated after 2011, when new mustard oil manufacturers from other Indian states, such as Rajasthan and Uttar Pradesh, entered the West Bengal market. These manufacturers had the advantage of being from some of the largest mustard-seed-producing states in the country. New, large companies with an established national presence in the food and edible oil industry also entered the mustard oil market in the branded and packaged segment, which was dominated by 10 to 12 companies. Prominent brands included Dhara (manufactured by Mother Dairy Fruit & Vegetable Pvt. Ltd.), Engine (Shree Hari Industries), Nutrela (Ruchi Soya Industries Limited), Fortune and Bullet brands (Adani Wilmar Limited), Patanjali (Patanjali Ayurved), and Kalash (K.S. Oil). These oil manufacturers and their mustard oil brands were the leaders in the mustard oil industry.

The majority of these large edible oil manufacturers sold a large variety of edible oils to capture the diverse preferences of edible oil consumers across the country. For instance, Adani Wilmar Limited had well-known brands in almost every edible oil category, including refined soybean oil under the brand name of Fortune, mustard oil under the brand name of Bullet, coconut oil under the brand name of Ivory, and refined palmolein oil under the brand name of Raag Gold. This was in sharp contrast to small manufacturers like HOI, which did not have manufacturing and processing capacity for more than one kind of edible oil. This strongly limited HOI's ability to serve diverse consumer preferences. However, HOI managed to compete with these large companies to a modest extent because of its positioning in Asansol as a supplier of pure mustard oil and its commitment to maintaining prices around 10 per cent lower than those of the large companies.

The remaining one-fifth of the market in Asansol was captured by small oil manufacturers, who competed strongly for minuscule market shares. HOI had dominated this category for decades, and this had helped it to maintain sales in a narrow and stable range of around 10,000 kg to 14,000 kg every month (see Exhibit 5).

CHALLENGES IN 2015

The year 2015 brought a set of unprecedented challenges for Abhishek. Lack of rainfall meant that mustard seed production in 2014–15 had dropped to 6.3 million tonnes—a five-year low[14] (see Exhibit 6). This led to a steep rise in the procurement price of mustard seeds for the oil mills. To cover the increased cost of production, almost all companies hiked the price of their mustard oil, and the retail price of mustard oil soared as a result. Smaller companies had no choice but to fully pass on the increased costs to their

[14] Sandip Das, "Now, Mustard Oil Prices Witness Sharp Increase," *The Financial Express*, October 30, 2015, accessed November 21, 2016, www.financialexpress.com/market/commodities/now-mustard-oil-prices-witness-sharp-increase/158701/.

consumers, while the larger companies could absorb this price shock because of economies of scale and efficient inventory management. Abhishek elaborated:

> Larger companies have bargaining power in terms of procurement, have large storage capacities for mustard seeds, and also use forward contracts.[15] They had procured seeds in bulk in the previous season when the price was low and had also utilized forwards to hedge oil seeds price risks. On the other hand, our mill had smaller storage capacity for oil seeds, which could support only 15 to 20 days of production of mustard oil. We also did not have any experience with forwards. Thus, we were nakedly exposed to the high price of mustard seeds.

HOI had never experimented with raising the price of Maa mustard oil closer to the prices of the large companies out of fear of losing its market share. However, the present situation compelled HOI to increase its price to cover increased costs. It was doubtful that HOI's larger competitors would also increase the price of their products by such a large amount. To get some idea about the expected increase in the competitors' prices, Abhishek held numerous discussions with the brokers and commission agents operating in the edible oil business. They expected that the competitors might increase their prices by around 6 per cent on average. However, the situation was fraught with uncertainties and demanded an analysis of the price-quantity relationship of Maa mustard oil with respect to the competitors' prices.

Such pricing dilemmas were not new to HOI because oil manufacturing operations were dependent on the production of oil seeds, which fluctuated depending on the quality of monsoons. However, this was the first time that such a situation had cropped up since Abhishek had taken control of the business operations. He did not want to continue with his father's strategy of maintaining the price differential even in such stressful conditions. The agricultural crop year of July 2011 to June 2012, when mustard seed production had fallen quite low, had exposed HOI to a similar plight. The final say in the decision-making at that point had been Ashok's, and he did not venture into pricing strategies. Instead, he had kept the prices almost static from July 2012 to March 2013, thereby maintaining his self-imposed price gap. As a result, HOI had to absorb high production costs, which significantly reduced its profits in the corresponding financial year. Abhishek was not in the mood to give up this time. He wanted to leverage HOI's established reputation to increase the price, irrespective of what the competitors did.

THE TASK AHEAD

Abhishek had to identify several factors that influenced the demand for Maa mustard oil and then incorporate the effects of all these factors to assess the demand for his product. His ultimate objective was to identify an optimum price for Maa mustard oil in these troubled times. He reflected on what he had learned in his managerial economics class and wondered if he could use some of those theories to resolve his present pricing dilemma. Could the price be raised without hurting revenues? Abhishek knew that the price hikes of competitors could not be perfectly estimated. What was the optimum price he could charge under various probable scenarios of competitors' price hikes?

Abhishek stared ahead without seeing a way forward. Although the future seemed grim, he was not about to give up. He set his jaw, rolled up his sleeves, and got down to work.

[15] Forward contracts were contracts between two parties to buy or sell assets at pre-specified prices on a future date; they could be used as an instrument for hedging against risks.

EXHIBIT 1: IMPORT DUTY ON RBD PALMOLEIN OIL WITH EFFECTIVE DATE

Effective Date	Aug. 11, 2006	Jan. 24, 2007	Apr. 13, 2007	Jul. 23, 2007	Mar. 21, 2008	Apr. 1, 2008	Jan. 20, 2014	Jan. 24, 2014	Sep. 17, 2015
Import Duty	80.0%	67.5%	57.5%	52.5%	27.5%	7.5%	10.0%	15.0%	20.0%

Source: Created by the authors based on “Sugar & Vegetable Oils,” Government of India, Department of Food and Public Distribution, accessed February 25, 2016, http://dfpd.nic.in/oil-division.htm.

EXHIBIT 2: MONTHLY PER CAPITA NET STATE DOMESTIC PRODUCT (NSDP) OF WEST BENGAL

Month and Year	Per capita NSDP (in ₹)	Month and Year	Per capita NSDP (in ₹)
April 2011	4,343.17	July 2013	5,691.44
May 2011	4,347.59	August 2013	5,757.95
June 2011	4,356.44	September 2013	5,821.98
July 2011	4,369.71	October 2013	5,883.52
August 2011	4,387.40	November 2013	5,942.57
September 2011	4,409.51	December 2013	5,999.14
October 2011	4,436.05	January 2014	6,053.22
November 2011	4,467.01	February 2014	6,104.82
December 2011	4,502.40	March 2014	6,153.93
January 2012	4,542.20	April 2014	6,200.56
February 2012	4,586.43	May 2014	6,252.15
March 2012	4,635.08	June 2014	6,308.70
April 2012	4,688.16	July 2014	6,370.21
May 2012	4,743.18	August 2014	6,436.68
June 2012	4,800.15	September 2014	6,508.11
July 2012	4,859.07	October 2014	6,584.51
August 2012	4,919.94	November 2014	6,665.86
September 2012	4,982.76	December 2014	6,752.18
October 2012	5,047.52	January 2015	6,843.45
November 2012	5,114.23	February 2015	6,939.69
December 2012	5,182.89	March 2015	7,040.89
January 2013	5,253.50	April 2015	7,147.05
February 2013	5,326.05	May 2015	7,244.36
March 2013	5,400.55	June 2015	7,332.83
April 2013	5,477.00	July 2015	7,412.45
May 2013	5,550.97	August 2015	7,483.22
June 2013	5,622.45	September 2015	7,545.15

Note: Monthly per capita NSDP was calculated using the Denton Cholette method of temporal disaggregation on annual per capita NSDP data of West Bengal; ₹ = INR = Indian rupee; US$1.00 = ₹65.57 on September 30, 2015.
Source: Created by the case authors based on Christoph Sax and Peter Steiner, “Temporal Disaggregation of Time Series,” *The R Journal* 5, no. 2 (2013): 80–87, accessed March 7, 2016, https://journal.r-project.org/archive/2013-2/sax-steiner.pdf; “Per Capita NSDP at Current Prices (2004–05 to 2014–15),” Government of India, National Institute for Transforming India Aayog, accessed March 13, 2016, http://niti.gov.in/content/capita-nsdp-current-prices-2004-05-2014-15.

EXHIBIT 3: COMPETITORS' AVERAGE PRICE OF EDIBLE MUSTARD OIL FROM APRIL 2011 TO SEPTEMBER 2015

Month and Year	Price per kg (in ₹)	Month and Year	Price per kg (in ₹)
April 2011	69.22	July 2013	95.00
May 2011	70.67	August 2013	92.72
June 2011	74.77	September 2013	95.00
July 2011	75.00	October 2013	95.25
August 2011	75.00	November 2013	96.53
September 2011	77.29	December 2013	98.00
October 2011	79.06	January 2014	97.89
November 2011	81.37	February 2014	93.47
December 2011	84.00	March 2014	95.30
January 2012	91.61	April 2014	93.76
February 2012	91.11	May 2014	92.06
March 2012	95.79	June 2014	92.00
April 2012	99.13	July 2014	92.00
May 2012	99.70	August 2014	92.00
June 2012	98.44	September 2014	94.29
July 2012	104.62	October 2014	95.29
August 2012	105.42	November 2014	98.11
September 2012	104.45	December 2014	100.00
October 2012	104.94	January 2015	101.06
November 2012	105.95	February 2015	101.10
December 2012	106.00	March 2015	99.40
January 2013	106.00	April 2015	99.22
February 2013	105.58	May 2015	101.58
March 2013	102.84	June 2015	107.95
April 2013	96.84	July 2015	108.52
May 2013	94.26	August 2015	107.76
June 2013	94.00	September 2015	109.14

Note: ₹ = INR = Indian rupee; US$1.00 = ₹65.57 on September 30, 2015; kg = kilogram.
Source: "Monthly Average Retail Prices of Mustard Oil (Packed)," Government of India, Department of Consumer Affairs, database, accessed March 9, 2016, http://fcainfoweb.nic.in/PMSver2/Reports/Report_Menu_web.aspx.

EXHIBIT 4: HISTORICAL DATA OF PROMOTIONAL EXPENDITURE INCURRED BY HIND OIL INDUSTRIES (IN INR)

Month and Year	Promotional Expenditure (in ₹)	Month and Year	Promotional Expenditure (in ₹)
April 2011	1,170.00	July 2013	1,084.50
May 2011	1,162.26	August 2013	1,096.92
June 2011	1,164.06	September 2013	1,080.63
July 2011	1,193.13	October 2013	1,069.38
August 2011	1,241.55	November 2013	1,069.47
September 2011	1,254.15	December 2013	1,039.14
October 2011	1,149.57	January 2014	932.04
November 2011	1,200.06	February 2014	1,052.37
December 2011	1,220.40	March 2014	970.11
January 2012	1,082.70	April 2014	943.47
February 2012	1,092.06	May 2014	1,043.37
March 2012	989.37	June 2014	1,050.03
April 2012	978.03	July 2014	1,056.51
May 2012	972.09	August 2014	1,048.14
June 2012	1,004.94	September 2014	993.69
July 2012	1,081.71	October 2014	1,004.22
August 2012	1,110.15	November 2014	961.38
September 2012	1,262.25	December 2014	869.13
October 2012	1,229.22	January 2015	814.68
November 2012	1,257.75	February 2015	801.99
December 2012	1,021.50	March 2015	868.68
January 2013	943.92	April 2015	925.11
February 2013	1,043.46	May 2015	991.26
March 2013	1,125.90	June 2015	956.79
April 2013	1,090.08	July 2015	1,123.20
May 2013	1,087.20	August 2015	1,202.13
June 2013	1,105.92	September 2015	1,247.31

Note: ₹ = INR = Indian rupee; US$1.00 = ₹65.57 on September 30, 2015.
Source: Company files.

EXHIBIT 5: SALES HISTORY OF MAA MUSTARD OIL FROM APRIL 2011 TO SEPTEMBER 2015

Month and Year	Price per kg (in ₹)	Quantity Sold (in kg)	Month and Year	Price per kg (in ₹)	Quantity Sold (in kg)
April 2011	65.00	12,890	July 2013	79.64	12,095
May 2011	65.90	12,850	August 2013	78.00	11,958
June 2011	65.40	13,180	September 2013	78.09	11,874
July 2011	65.34	13,785	October 2013	78.14	11,850
August 2011	66.00	13,880	November 2013	81.54	11,535
September 2011	69.00	12,680	December 2013	81.27	10,258
October 2011	69.00	13,290	January 2014	80.01	11,680
November 2011	69.25	13,498	February 2014	80.00	10,768
December 2011	74.56	11,980	March 2014	80.81	10,457
January 2012	78.00	12,085	April 2014	78.98	11,527
February 2012	84.00	10,980	May 2014	78.76	11,615
March 2012	87.00	10,820	June 2014	78.23	11,675
April 2012	90.00	10,768	July 2014	78.25	11,600
May 2012	90.00	11,125	August 2014	77.88	10,980
June 2012	88.00	11,985	September 2014	77.34	11,099
July 2012	89.00	12,248	October 2014	78.59	10,657
August 2012	89.00	13,985	November 2014	83.53	9,645
September 2012	92.00	13,580	December 2014	87.29	9,026
October 2012	91.00	13,895	January 2015	90.72	8,875
November 2012	94.00	11,270	February 2015	88.03	9,585
December 2012	95.00	10,485	March 2015	85.63	10,256
January 2013	91.00	11,545	April 2015	84.25	10,985
February 2013	86.00	12,465	May 2015	87.90	10,542
March 2013	86.00	12,025	June 2015	90.01	12,465
April 2013	85.00	11,985	July 2015	90.00	13,275
May 2013	80.00	12,245	August 2015	88.89	13,786
June 2013	80.01	11,976	September 2015	91.38	13,256

Note: ₹ = INR = Indian rupee; US$1.00 = ₹65.57 on September 30, 2015; kg = kilogram.
Source: Company files.

EXHIBIT 6: ANNUAL PRODUCTION OF MUSTARD SEEDS FROM 2010–11 TO 2014–15

Year	2010–11	2011–12	2012–13	2013–14	2014–15
Mustard Seed Production (in million tonnes)	8.1	6.6	8.0	7.8	6.3

Source: Sandip Das, "Now, Mustard Oil Prices Witness Sharp Increase," *The Financial Express*, October 30, 2015, accessed February 21, 2016, www.financialexpress.com/market/commodities/now-mustard-oil-prices-witness-sharp-increase/158701/.